

Enhanced Cerebrovascular Responsiveness to Hypercapnia Following Depletion of Central Serotonergic Terminals

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Summary: Serotonin-containing nerve fibres innervate cerebral blood vessels, but the source of this innervation and the physiological effects of perivascular serotonin release remain controversial. The purpose of the present study was to examine the effects of central serotonergic depletion upon the relationship between CBF and glucose utilization under both normo- and hypercapnic conditions. To induce the loss of serotonergic terminals, rats were injected twice daily for 4 consecutive days with 20 mg/kg of the specific serotonergic neurotoxin methylenedioxymphetamine (MDA). Between 4 and 6 weeks later, local CBF and glucose utilization were measured using the fully quantitative [^{14}C]iodoantipyrine and [^{14}C]2-deoxyglucose autoradiographic techniques, respectively, and the efficacy of the lesioning protocol was assessed using [^3H]paroxetine radioligand binding analysis. In all animals treated with MDA, there was a significant decrease in serotonin uptake sites throughout the brain, falling from 223 ± 20 to 40 ± 16 fmol/mg tissue in parietal cortex, for example, although the raphe nuclei them-

selves were unaffected (300 ± 20 fmol/mg tissue in controls and 291 ± 18 in MDA-treated rats). In normocapnic rats, the effects of MDA pretreatment upon blood flow and glucose use were slight and focally concentrated. However, when the animals were rendered hypercapnic, CBF was significantly higher in MDA-treated rats than in normal controls, for example, increasing from 356 ± 22 ml $100 \text{ g}^{-1} \text{ min}^{-1}$ in frontal cortex of hypercapnic controls to 700 ± 81 ml $100 \text{ g}^{-1} \text{ min}^{-1}$ in MDA-pretreated rats with similar levels of hypercapnia. In some brain areas of hypercapnic MDA-pretreated rats, blood flows were too high (>800 ml $100 \text{ g}^{-1} \text{ min}^{-1}$) to be accurately quantified. The results of this study are consistent with a constrictor role for perivascular serotonergic innervation that becomes manifest when, as in hypercapnia, cerebrovascular tone is low. **Key Words:** Cerebral blood flow—Cerebral glucose utilization—5-Hydroxytryptamine—Hypercapnia—Methylenedioxymphetamine—Paroxetine binding.

There is now compelling evidence that cerebral blood vessels are innervated by serotonergic neurones arising from mesencephalic sites within the brain (Steinbusch, 1981; Edvinsson et al., 1983;

Scatton et al., 1985; Bonvento et al., 1991). These findings are not, however, uncontroversial (Cowan et al., 1987; Jackowski et al., 1989), and the issue of serotonin (5-HT) uptake by perivascular sympathetic nerves originating from the superior cervical ganglia continues to fuel debate (Moreno et al., 1994). Whatever the source of cerebrovascular serotonergic innervation, 5-HT and its agonists are known to be profoundly vasoactive (Edvinsson et al., 1978; Forster and Whalley, 1981, 1982; Fu and Toda, 1983), and 5-HT receptors have been identified on large cerebral arteries and pial arterioles (Müller-Schweinitzer and Engel, 1983; Peroutka et al., 1983; Hamel et al., 1989). The concept of an intrinsic central serotonergic effector mechanism by which changes in CBF may be coupled to alterations in neuronal function may be an attractive

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Abbreviations used: LCBF, local CBF; LCMR_{glc}, local CMR_{glc}; 5-HT, 5-hydroxytryptamine; MDA, methylenedioxymphetamine; MDMA, methylenedioxymethamphetamine.

proposition (Lou et al., 1987), but as yet there is no clear physiological evidence for such a role.

Despite the apparently widespread distribution of perivascular serotonergic innervation of central origin (Duverger et al., 1987; Bonvento et al., 1991), intracerebral interventions to either stimulate (Bonvento et al., 1989; Cudennec et al., 1989) or induce specific serotonergic lesions (Dahlgren et al., 1981; McBean et al., 1990a; Underwood et al., 1992) have failed to provide a clear indication of the role played by 5-HT neurones in the control of CBF. Even when cerebrovascular tone is low (as in hypercapnia), and it might be anticipated that the loss of endogenous constrictor effects of 5-HT might be more clearly manifest (Rosenblum and Nelson, 1990), the results are contradictory (Itakura et al., 1985; Underwood et al., 1992). The reasons for these inconsistencies are not clear but may lie in differences in measurement technique or anaesthetic regimens. The purpose of the present study was to apply quantitative autoradiographic techniques to the investigation of effects of hypercapnia upon local cerebral blood flow (LCBF) and glucose utilization (LCMR_{glc}) in the conscious rat brain following specific depletion of central serotonergic terminals induced by the amphetamine derivative methylenedioxyamphetamine (MDA).

METHODS

Animal treatment protocols

A total of 40 male Sprague-Dawley rats, weighing ~50 g at the outset, were used in this study. The amphetamine derivative ($\pm$)-3,4-methylenedioxyamphetamine hydrochloride (MDA; Sigma) was dissolved in saline, and one group of rats ($n = 20$) were injected subcutaneously twice daily for 4 days with 20 mg/kg in a volume of 0.25 ml. Control rats ($n = 20$) were injected with saline alone. The rats were held under normal animal house conditions with free access to food and water for 4–6 weeks before the measurement of LCBF or LCMR_{glc}.

Animal preparation

On the day of the experiment, the animals (now weighing 250–300 g) were anaesthetised with halothane (maintained at 1% in a mixture of 70% nitrous oxide/30% oxygen), and polythene cannulae were inserted into both femoral arteries, to allow sampling of arterial blood and the measurement of arterial blood pressure, and both femoral veins, for the injection of radiolabelled tracers and barbiturate at the time of killing. Following the application of a loose-fitting plaster cast to the pelvis and hind limbs, the rats were supported on a weighted platform to which the plaster was secured with adhesive tape and were placed into a polythene dome into which air was piped at a rate of 1.5 L/min. The rats were allowed to recover from the effects of the anaesthesia for at least 2 h before any further experimental manipulation, during

which arterial pressure and rectal temperature were monitored continuously. At the end of this recovery period, hypercapnia was induced in equal numbers of MDA- and saline-pretreated rats ($n = 10$ from each treatment group) by the introduction of carbon dioxide into the air supply until arterial PCO₂ tensions within the range of 55–65 mm Hg were attained. Normocapnic animals ($n = 10$ from each treatment group) continued to be supplied with air alone. Following a 10-min equilibrium period and at 5-min intervals over the subsequent 20 min, samples of arterial blood (80 μ l) were withdrawn for gas analysis (Corning 168 Gas Analyzer) to ensure constancy of CO₂ tensions. The measurement of LCMR_{glc} was initiated at the end of the 10-min equilibrium period, whilst LCBF studies were initiated after 20 min. The time differential was chosen so that the measurement of LCBF coincided approximately with the median point of integrated specific 2-deoxyglucose in the brain. For each of the LCBF and LCMR_{glc} measurements, there were four different treatment groups, each with $n = 5$: normocapnia + saline pretreatment; normocapnia + MDA pretreatment; hypercapnia + saline pretreatment; and hypercapnia + MDA pretreatment.

Measurement of LCBF

CBF was measured in 20 rats (5 from each treatment group) using the quantitative [¹⁴C]iodoantipyrine autoradiographic technique (Sakurada et al., 1978). The tracer (50 μ Ci/rat in 0.6 ml saline) was infused intravenously at a constantly accelerating rate over 45 s. Over the infusion period, blood was allowed to flow freely from one of the arterial cannulae and samples were collected intermittently onto preweighed filter discs that were reweighed after the experiment to determine the sample size. Subsequently, the timing of each sample was determined precisely from an audiocassette recording of the experiment and corrected for the flow characteristics of the cannula tubing. At 45 s, the animals were killed by the rapid intravenous injection of sodium pentobarbitone, and the brains were dissected intact and frozen in precooled 2-methylbutane (-45°C) within 2–3 min of death. The filter discs were placed into scintillation vials and 0.4 ml hydrogen peroxide added to bleach the pigment prior to liquid scintillation analysis.

Measurement of LCMR_{glc}

Glucose use was measured in 20 rats (5 from each treatment group) using the fully quantitative [¹⁴C]2-deoxyglucose autoradiographic technique (Sokoloff et al., 1977). The measurement was initiated with a 30-s intravenous injection of tracer (50 μ Ci/rat in 0.75 ml saline). Over the subsequent 45 min, a total of 14 timed arterial blood samples were collected at predetermined intervals. These samples were centrifuged to separate plasma from the cellular fraction. Aliquots of each plasma sample were taken for the determination of ¹⁴C concentrations (20 μ l) and glucose levels (10 μ l) by liquid scintillation analysis and semiautomated glucose oxidase assay (Beckman), respectively. At the end of the measurement period, the rats were killed and the brains prepared as for blood flow experiments.

Preparation and analysis of autoradiograms

Cryostat sections (20 μ m) were cut in the coronal plane from each brain. Three consecutive sections were collected from every 200 μ m throughout the brain from me-

dulla to frontal cortex, mounted onto glass coverslips, and rapidly dried on a hotplate. Sections adjacent to those used for autoradiographic measurement of LCMR_{glc} and LCBF were thaw-mounted onto gelatin-covered glass slides and stored at -70°C for subsequent [^3H]paroxetine autoradiographic binding analysis of serotonergic (5-HT) uptake sites. The slide-mounted sections were prepared for [^3H]paroxetine autoradiography according to the protocol described previously (Battaglia et al., 1987) with the addition of preliminary wash procedures designed to remove ^{14}C tracers from the tissue (Chalmers and McCulloch, 1989). Sections were then incubated in buffer containing a saturating concentration (250 pM) of [^3H]paroxetine. Nonspecific binding was defined in adjacent sections by [^3H]paroxetine binding in the presence of 4 μM citalopram.

Analysis of [^{14}C]2-deoxyglucose, [^{14}C]iodoantipyrine, and [^3H]paroxetine autoradiograms was performed using a computer-based image analysis system (Cambridge Instruments Quantimet 970). Tissue isotope concentrations were measured from autoradiographic images of brain sections relative to appropriate ^{14}C - or ^3H -containing standards (Amersham International) and LCMR_{glc} or LCBF calculated using the operational equations for the techniques (Sokoloff et al., 1977; Sakurada et al., 1978). Measurements were taken from 25 functionally diverse brain regions from each of the perfusion territories supplied by the major cerebral arteries. Specific [^3H]paroxetine binding was determined by subtraction of image densities from total and nonspecific binding images.

Statistical analysis

Data were evaluated statistically by analysis of variance followed by post hoc Scheffé test to allow multiple comparisons. Acceptable levels of significance were set at $p < 0.05$. Some of the regional blood flow values generated in this study were above the limits of accurate quantification for the iodoantipyrine technique and were reported as $>800 \text{ ml } 100 \text{ g}^{-1} \text{ min}^{-1}$. This resulted in an incomplete data set that precluded the use of the rigorous mode of statistical analysis of the flow-metabolism relationship developed for experiments of this design (McCulloch et al., 1982). To determine whether there

were any differences in the overall relationship of LCBF to LCMR_{glc} between treatment groups, we therefore applied the Mann-Whitney U test to the ratios of mean LCBF to mean LCMR_{glc} for each of the brain areas.

RESULTS

Physiological variables

There were no significant differences in physiological variables between saline- and MDA-pretreated groups under either normocapnic or hypercapnic conditions (Table 1), but within each treatment group, arterial PCO_2 and pH were significantly increased and decreased, respectively, in rats breathing air supplemented with CO_2 . Arterial PO_2 was also significantly increased in these animals, most probably due to the increased respiratory effort that accompanied the hypercapnia. Neither plasma glucose levels nor arterial blood pressures were significantly affected by either MDA pretreatment or hypercapnia, suggesting that levels of stress in these animals were no higher than in controls.

Paroxetine binding autoradiography

The effects of MDA treatment upon the distribution of [^3H]paroxetine-labelled 5-HT uptake sites were clearly evident from visual inspection of the autoradiographic images. In control animals, [^3H]paroxetine binding was found throughout the brain, with highest densities evident in neocortex, hippocampus, and elements of the extrapyramidal system (Table 2). In keeping with our previously published observations (McBean et al., 1990a,b), repeated exposure to MDA resulted in widespread and profound reductions in [^3H]paroxetine binding. In neocortex, for example, binding densities (B_{max}) de-

TABLE 1. Physiological variables in MDA-pretreated rats

Experimental groups	Physiological variables					
	PCO_2 (mm Hg)	PO_2 (mm Hg)	pH	Plasma glucose (g/L)	MABP (mm Hg)	Temperature ($^{\circ}\text{C}$)
Saline/normocapnia						
LCMR_{glc}	40.8 ± 1.3	86.0 ± 10.1	7.42 ± 0.02	1.42 ± 0.11	131 ± 10	37.1 ± 0.4
LCBF	38.4 ± 1.1	85.2 ± 8.3	7.41 ± 0.04	1.38 ± 0.04	122 ± 7	37.2 ± 0.5
Saline/hypercapnia						
LCMR_{glc}	61.2 ± 2.2^a	112.5 ± 5.1^a	7.27 ± 0.01^a	1.36 ± 0.20	122 ± 9	37.0 ± 0.2
LCBF	62.2 ± 2.4^a	101.0 ± 7.6^a	7.30 ± 0.03^a	1.49 ± 0.08	116 ± 6	36.9 ± 0.4
MDA/normocapnia						
LCMR_{glc}	42.0 ± 2.2	87.9 ± 8.0	7.42 ± 0.02	1.68 ± 0.16	120 ± 9	36.9 ± 0.2
LCBF	37.9 ± 4.2	91.1 ± 3.8	7.43 ± 0.04	1.57 ± 0.13	121 ± 8	37.2 ± 0.2
MDA/hypercapnia						
LCMR_{glc}	57.5 ± 5.6^a	113.0 ± 3.6^a	7.29 ± 0.03^a	1.56 ± 0.11	126 ± 3	37.0 ± 0.7
LCBF	62.3 ± 2.5^a	113.0 ± 3.4^a	7.28 ± 0.02^a	1.68 ± 0.16	121 ± 9	36.8 ± 0.2

All data are presented as means $\pm$ SD. For each measurement group, $n = 5$. For abbreviations see text.

^a Significantly different from appropriate normocapnic measurement group, $p < 0.05$, an analysis of variance with post hoc Scheffé test for multiple comparisons. There were no significant differences between saline and MDA treatment groups under either normo- or hypercapnic conditions.

TABLE 2. Effects of MDA upon [³H]paroxetine binding in rat brain

Brain region	Pretreatment	
	Saline	MDA
Raphe nuclei		
Dorsal	300 ± 20	291 ± 18
Medial	159 ± 15	169 ± 10
Inferior colliculus	200 ± 16	17 ± 9 ^a
Visual cortex	98 ± 9	28 ± 8 ^a
Auditory cortex	65 ± 6	32 ± 8 ^a
Medial geniculate	48 ± 7	19 ± 12 ^a
Hippocampus		
Molecular layer	95 ± 10	42 ± 12 ^a
Dentate gyrus	49 ± 8	38 ± 6 ^a
Lateral geniculate	350 ± 18	56 ± 13 ^a
Parietal cortex	223 ± 20	40 ± 16 ^a
Caudate nucleus	98 ± 12	15 ± 8 ^a
Sensorimotor cortex	120 ± 14	24 ± 15 ^a
Nucleus accumbens	95 ± 18	21 ± 12 ^a
Cingulate cortex	139 ± 15	14 ± 9 ^a
Frontal cortex	64 ± 8	8 ± 4 ^a

All data are presented as mean binding density (fmol/mg tissue) ± SD. For each measurement group, n = 10. For abbreviation see text.

^a Significantly different from saline pretreatment group, p < 0.05 (Student's *t* test).

creased by between 50 and 90% following MDA treatment. In a few other areas, most notably in the raphe nuclei, binding levels were largely preserved (Table 2).

LCBF

In keeping with previously published data (McBean et al., 1990a), MDA pretreatment produced focal increases in LCBF (Table 3). Hypercapnia in saline-pretreated rats produced global increases in blood flow, which were statistically significant in all but two brain regions (flocculus and median raphe) (Table 3). Similar global increases were evident in hypercapnic MDA-pretreated rats, but in the majority of areas (20 of 25), LCBF was significantly higher than in saline-treated rats with similar levels of hypercapnia (Table 3). Indeed, in every MDA-pretreated rat, hypercapnia produced blood flows that in some areas were so high that it proved impossible to quantify them accurately. In these regions, the values reported (800 ml 100 g⁻¹ min⁻¹) may be an underestimate (Table 3).

LCMR_{glc}

In saline-pretreated rats, hypercapnia resulted in significant decreases in mean LCMR_{glc} in 4 of the 25 brain areas analysed (Table 4). These brain regions were found exclusively within the motor system (cerebellum, striatum, and motor cortex), which is consistent with the reduced motor activity observed in these hypercapnic rats. Although a more general pattern of metabolic depression was evident in hy-

percapnic rats previously exposed to MDA (Table 4), this was against a pattern of widespread, though largely nonsignificant, increases in LCMR_{glc} under normocapnic conditions. Apart from two areas (inferior colliculus and visual cortex) in which the MDA-induced increase in LCMR_{glc} was maintained with hypercapnia, there were no significant differences in LCMR_{glc} between hypercapnic saline- and MDA-pretreated rats (Table 4).

Relationship between LCBF and LCMR_{glc}

In saline-treated rats, the ratio of LCBF to LCMR_{glc} increased in every area of the brain under hypercapnic conditions (Table 5), and over the brain as a whole, the ratios (median ratio = 4.38) were significantly greater than in normocapnic rats (median ratio = 1.97) (p = 0.00, Mann-Whitney *U* test). However, in the MDA-treated group under hypercapnic conditions, ratios of LCBF to LCMR_{glc} were higher (median ratio = 6.00) than they were in intact hypercapnic controls, the only exceptions being the copula pyramis and white matter of the cerebellum and cingulate cortex (Table 5). Statistical analysis revealed that over the brain as a whole, the difference in ratios between intact and lesioned animals under hypercapnic conditions was highly significant (p = 0.0003, Mann-Whitney *U* test).

DISCUSSION

Although the neurotoxic properties of MDA have yet to be fully elucidated, it is clearly evident from immunohistochemical, neurochemical, and radioligand binding studies that repeated exposure to MDA induces highly specific depletion of serotonergic terminals in the rat brain whilst having no detectable effects upon either dopaminergic or noradrenergic terminals (Battaglia et al., 1987). Neurochemical indices of serotonergic terminal depletion induced by MDA (decreased 5-HT content and metabolite concentrations) are paralleled both qualitatively and quantitatively by changes in the densities of [³H]paroxetine-labelled 5-HT uptake sites (see Fig. 1), and in the present study reductions of ~70% in paroxetine binding are similar to those reported previously using the same treatment regimen (Battaglia et al., 1987; McBean et al., 1990a,b). Whilst we were thus able to confirm the efficacy of our lesion relative to previously published accounts, with all the consequences for cerebral 5-HT systems which that entails, we did not have any direct measure of alterations in perivascular serotonergic processes. Although there is no *a priori* reason to expect that perivascular serotonergic fibres should react any different from those in the neuropil, in the

TABLE 3. Effects of hypercapnia upon LCBF in saline- and MDA-pretreated rats

Brain region	Normocapnia		Hypercapnia	
	Saline	MDA	Saline	MDA
Cerebellum				
Cerebellar pyramis	109 ± 4	144 ± 12	252 ± 22 ^b	285 ± 25 ^b
Paramedian lobule	91 ± 11	93 ± 5	153 ± 19 ^b	192 ± 18 ^{a,b}
Flocculus	190 ± 20	323 ± 71	438 ± 69	>800 ^{a,b}
White matter	41 ± 4	44 ± 5	85 ± 8 ^b	81 ± 12 ^b
Raphe nuclei				
Dorsal	147 ± 9	370 ± 22 ^a	361 ± 40 ^b	>800 ^{a,b}
Medial	142 ± 4	217 ± 17	276 ± 21	>800 ^{a,b}
Inferior colliculus	258 ± 13	460 ± 25 ^a	541 ± 49 ^b	>800 ^{a,b}
Visual cortex	198 ± 13	260 ± 20	338 ± 15 ^b	690 ± 94 ^{a,b}
Auditory cortex	320 ± 31	450 ± 12	509 ± 99 ^b	>800 ^{a,b}
Medial geniculate	204 ± 13	288 ± 8	519 ± 73 ^b	>800 ^{a,b}
Hippocampus				
Molecular layer	148 ± 11	141 ± 9	334 ± 53 ^b	340 ± 20 ^b
Dentate gyrus	122 ± 11	115 ± 12	276 ± 37 ^b	290 ± 20 ^b
Subthalamic nucleus	152 ± 22	184 ± 13	324 ± 48 ^b	530 ± 30 ^{a,b}
Lateral geniculate	148 ± 9	172 ± 14	334 ± 66 ^b	471 ± 42 ^{a,b}
Ventromedial thalamus	198 ± 12	198 ± 12	456 ± 41 ^b	566 ± 45 ^{a,b}
Medial habenula	163 ± 16	197 ± 21	435 ± 90 ^b	605 ± 60 ^{a,b}
Parietal cortex	225 ± 29	252 ± 16	455 ± 79 ^b	702 ± 88 ^{a,b}
Globus pallidus	75 ± 4	96 ± 10	162 ± 20 ^b	210 ± 30 ^{a,b}
Corpus callosum	43 ± 4	45 ± 3	80 ± 6 ^b	120 ± 8 ^{a,b}
Caudate nucleus	137 ± 11	157 ± 10	295 ± 47 ^b	402 ± 30 ^{a,b}
Sensorimotor cortex	260 ± 34	291 ± 9	484 ± 80 ^b	786 ± 94 ^{a,b}
Nucleus accumbens	175 ± 9	161 ± 12	323 ± 46 ^b	502 ± 80 ^{a,b}
Cingulate cortex	224 ± 18	249 ± 16	572 ± 53 ^b	620 ± 59 ^b
Frontal cortex	245 ± 20	267 ± 22	356 ± 22 ^b	700 ± 81 ^{a,b}
Genu	49 ± 9	49 ± 6	77 ± 6 ^b	118 ± 11 ^{a,b}

All data are presented as mean LCBF (ml 100 g⁻¹ min⁻¹) ± SD. For each measurement group, n = 5. For abbreviations see text.

^a Significantly different from appropriate saline-pretreatment group.

^b Significantly different from similarly pretreated normocapnic group, p < 0.05 (analysis of variance followed by post hoc Scheffé test for multiple comparisons).

absence of direct evidence, the possibility cannot be completely discounted, and any attempt to correlate changes in local blood flow with changes in gross regional paroxetine binding must be treated with caution. Moreover, the paroxetine binding in this study will reflect to a large extent intraparenchymal terminal loss, and it is not surprising therefore that no such correlation appears to exist. This is most amply exemplified in the dorsal raphe nucleus where paroxetine binding was unaffected by MDA pretreatment and yet blood flow under both normo- and hypercapnic conditions was significantly higher than control.

Numerous studies have shown 5-HT to be a potent cerebrovascular constrictor, both in vitro (Toda et al., 1976; Forster and Whalley, 1982; Fu and Toda, 1983) and in situ (Edvinsson et al., 1977). Although these studies examined large cerebral arteries and pial arterioles, following circumvention of the blood-brain barrier, 5-HT administered systemically produces reductions in CBF in the whole animal (Harper and MacKenzie, 1977; Grome and Harper, 1983), which would be consistent with

there being a similar serotonergic constrictor influence upon the resistance vessels. More recently, pharmacological inhibition of endogenous 5-HT release, by systemic application of an agonist specific for the 5-HT_{1A} autoreceptor, was found to induce increases in LCBF that were in excess of any concomitant changes in LCMR_{glc} in the physiologically normal rat (McBean et al., 1991). Moreover, localized inhibition of 5-HT release with intracerebral injections of a 5-HT_{1B} agonist also increases CBF and produces focal flow-metabolism uncoupling (Kelly et al., 1987). Taken together, the results of these pharmacological studies are consistent with a perivascular serotonergic innervation providing vasoconstrictor tone throughout the cerebrovascular tree, from the major arteries to the intraparenchymal resistance vessels.

Although 5-HT may be taken up by perivascular sympathetic nerves innervating the large cerebral vessels (Saito and Lee, 1987), it is not these that provide the principal resistance elements to the cerebrovascular bed. This role is provided by the small arteries and arterioles, and there is evidence

TABLE 4. Effects of hypercapnia upon LCMR_{glc} in saline- and MDA-pretreated rats

Brain region	Normocapnia		Hypercapnia	
	Saline	MDA	Saline	MDA
Cerebellum				
Cerebellar pyramis	73 ± 7	83 ± 10	50 ± 15 ^b	59 ± 6 ^b
Paramedian lobule	54 ± 4	57 ± 6	45 ± 8	46 ± 8
Flocculus	99 ± 9	122 ± 12 ^a	64 ± 12 ^b	88 ± 9 ^{a,b}
White matter	34 ± 7	35 ± 5	30 ± 5	38 ± 7
Raphe nuclei				
Dorsal	74 ± 9	90 ± 7	72 ± 12	65 ± 9 ^b
Medial	76 ± 7	97 ± 10 ^a	63 ± 13	70 ± 6 ^b
Inferior colliculus	157 ± 11	177 ± 15	152 ± 11	185 ± 14 ^a
Visual cortex	104 ± 9	119 ± 5 ^a	101 ± 7	118 ± 10 ^a
Auditory cortex	125 ± 7	180 ± 10	131 ± 12	150 ± 16 ^b
Medial geniculate	109 ± 7	120 ± 9	102 ± 9	117 ± 9
Hippocampus				
Molecular layer	72 ± 7	96 ± 6 ^a	76 ± 6	67 ± 8 ^b
Dentate gyrus	60 ± 8	73 ± 8	64 ± 10	64 ± 12
Subthalamic nucleus	66 ± 7	73 ± 4	62 ± 7	72 ± 9
Lateral geniculate	64 ± 10	74 ± 7	64 ± 11	75 ± 8
Ventromedial thalamus	109 ± 10	109 ± 10	108 ± 12	117 ± 10
Medial habenula	79 ± 8	87 ± 11	74 ± 8	81 ± 5
Parietal cortex	101 ± 12	115 ± 12	94 ± 9	101 ± 8
Globus pallidus	38 ± 4	48 ± 5 ^a	38 ± 5	40 ± 3 ^b
Corpus callosum	27 ± 3	26 ± 4	25 ± 4	20 ± 5
Caudate nucleus	94 ± 8	111 ± 9 ^a	73 ± 5 ^b	80 ± 6 ^b
Sensorimotor cortex	110 ± 11	125 ± 8 ^a	92 ± 4 ^b	108 ± 5 ^b
Nucleus accumbens	71 ± 4	78 ± 7	61 ± 10	61 ± 4 ^b
Cingulate cortex	97 ± 6	135 ± 8 ^a	87 ± 7	95 ± 8 ^b
Frontal cortex	107 ± 8	128 ± 1 ^a	105 ± 6	112 ± 7 ^b
Genu	25 ± 6	30 ± 7	25 ± 8	20 ± 3

All data are presented as mean LCMR_{glc} (μmol 100 g⁻¹ min⁻¹) ± SD. For each measurement group, n = 5. For abbreviations see text.

^a Significantly different from appropriate saline-pretreatment group.

^b Significantly different from similarly pretreated normocapnic group, p < 0.05 (analysis of variance followed by post hoc Scheffé test for multiple comparisons).

that the serotonergic nerve fibres associated with these vessels arise from the raphe nuclei (Edvinsson et al., 1983; Scatton et al., 1985; Bonvento et al., 1991). Given the pharmacological evidence for a tonic influence of this central serotonergic innervation upon the cerebral circulation, it is surprising that selective toxins directed at serotonergic nerves appear not to alter resting CBF to any great extent (McBean et al., 1990a), if at all (Dahlgren et al., 1981; Underwood et al., 1992). The results of the present study do, however, indicate that serotonergic nerves may have a role to play in the regulation of LCBF under hypercapnic conditions and would be consistent with an intact serotonergic system providing a constrictor influence that moderates cerebrovascular dilatation induced by hypercapnia. Such a limited physiological role for 5-HT would be in keeping with the constrictor influence of 5-HT being most pronounced in vessels with low tone (Edvinsson et al., 1977; Rosenblum and Nelson, 1990), as will be the case with hypercapnia.

Although the results of this study are in keeping with previous measures of cortical blood flow fol-

lowing serotonergic lesions (Itakura et al., 1985), they are clearly not in agreement with the long-standing observations of Dahlgren et al. (1981), who showed that 5,7-DHT-induced serotonergic lesions had no effect upon hypercapnia-induced cerebrovascular dilatation, or the more recent results from Underwood et al. (1992) using electrolytic lesions. However, those studies were performed upon rats that were anaesthetised, paralysed, and artificially ventilated. The importance of anaesthetic effects in studies of this type has been highlighted recently (Edvinsson et al., 1993), and these factors must represent a real concern in the interpretation of conflicting data.

In these studies we have provided evidence that hypercapnia-induced increases in blood flow are enhanced by the loss of 5-HT neurones, but the effect upon the flow-metabolism relationship arises from two contributory factors. Whilst hypercapnic LCBF in MDA-treated rats is higher than in saline-treated rats, LCMR_{glc} is also decreased in a substantial proportion of the brain regions analysed, albeit to similar levels following both saline and

TABLE 5. Effects of hypercapnia upon flow-metabolism ratios in saline- and MDA-pretreated rats

Brain region	Normocapnia		Hypercapnia	
	Saline	MDA	Saline	MDA
Cerebellum				
Copula pyramis	1.49	1.73	5.04	4.83
Paramedian lobule	1.69	1.63	3.40	4.17
Flocculus	1.92	2.65	6.84	>9.09
White matter	1.21	1.26	2.83	2.13
Raphe nuclei				
Dorsal	1.99	4.11	5.01	>12.30
Medial	1.87	2.24	4.38	>11.42
Inferior colliculus	1.64	2.60	3.56	>4.32
Visual cortex	1.90	2.18	3.35	5.84
Auditory cortex	2.56	2.50	3.89	>5.33
Medial geniculate	1.87	2.40	5.09	>6.83
Hippocampus				
Molecular layer	2.06	1.45	4.39	5.07
Dentate gyrus	2.03	1.56	4.31	4.53
Subthalamic nucleus	2.30	2.52	5.23	7.36
Lateral geniculate	2.31	2.32	5.22	6.28
Ventromedial thalamus	1.82	1.82	4.22	4.84
Medial habenula	2.06	2.03	5.88	7.47
Parietal cortex	2.23	2.19	4.84	6.95
Globus pallidus	1.97	2.00	4.26	5.25
Corpus callosum	1.59	1.73	3.20	6.00
Caudate nucleus	1.46	1.41	4.04	5.03
Sensorimotor cortex	2.36	2.33	5.26	7.28
Nucleus accumbens	2.46	2.06	5.30	8.23
Cingulate cortex	2.31	1.84	6.57	6.53
Frontal cortex	2.29	2.09	3.39	6.25
Genu	1.96	1.63	3.08	5.90

Data are derived from the ratios of mean LCBF to mean LCMR_{glc} for each of the regions in each treatment group. For abbreviations see text.

MDA pretreatment. The literature on the metabolic response of the brain to hypercapnia is somewhat confused, with either no change (Eklöf et al., 1973; Artu and Michenfelder, 1980) or increased cerebral metabolism being reported (Berntman et al., 1979). Although in this study hypercapnia-induced decreases in LCMR_{glc} were observed in intact animals, the effects were limited to areas involved in motor function and are likely to have been secondary to the overall reduction in general motor activity, which, in these conscious animals, was observed to accompany an increased respiratory effort.

The results of this study have clinical implications that may go beyond the sphere of cerebrovascular research. The related amphetamine derivative methylenedioxymethamphetamine (MDMA; "Ecstasy") is a recreational drug used on a regular basis by growing numbers of individuals, and preliminary studies have indicated that the effects of MDA identified in this study are also shared by MDMA (Sharkey et al., 1993). The loss of cerebrovascular tone when an individual is physiologically challenged may have profound pathophysiological conse-



FIG. 1. Autoradiographic images from coronal sections of rat brain at the level of the medial geniculate bodies illustrating the distributions of [³H]paroxetine-labelled 5-HT uptake sites in a control rat (top) and in a rat pretreated with MDA 4 weeks previously (bottom). Note the contrast between high image densities in neocortex and hippocampus of control brain when compared with brain exposed to MDA, but also the relative sparing of binding in the nigral region of treated brain. For abbreviations see text.

quences and could be a contributory factor in the aetiology of cerebrovascular accidents reported from abusers of MDMA (Harries and DeSilva, 1992).

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